Algorithm K-index calculations

This document describes the algorithm developed by Dr. Ingemar Häggström to compute local K-indices for Kiruna (IAGA Code: KIR) magnetometer.

Definition K-index: Quantifies disturbances in the horizontal components (x and y) of the magnetic field in the range 0-9.

The index is computed as the difference between the min and max value for x och y component in three hours intervals. The max and min values are the differences between the component value and the presumed solar quiet value.

The values that are used in Kiruna region are the following:

K-indices	**Deflection in nT**
0	0-15
1	15-30
2	30-60
3	60-120
4	120-210
5	210-360
6	360-600
7	600-990
8	990-1500
9	>1500

Table 1

To be able to calculate the K-indices the solar quiet curve for the x- och y-component must be computed. The solar quiet curve is defined as the curve when the solar activity is low and it has shown that the geomagnetic field on the ground shows regular variation with a fundamental period of 24 h during a solar quiet day. Therefore the solar quiet curve consists of 24 values, one for each hour during a day. The curve is updated every month according to the following algorithm:

1. Read in the 24 solar quiet mean values for a day from previous month (in table 2 you can see an example of a solar quiet file).
x, y and z means / hour (24 values).

2. Calculate mean values for every second from **1.** by interpolating.

3. Calculate new mean values by first doing a solar quiet interval file (in table 3 you can see an example of such an interval file). That means get the time intervals that has values that are qualified to participate in the calculations of the new solar quiet file. Those intervals are produced by comparing the values of the file from **2.** with the value for the corresponding time of day for the date in question. The difference between those values must be within 100 nT (K-index 3) for that time to be qualified to participate in the calculation of the new solar quiet mean values.

4. Calculate new mean values from the file that is generated from **3.** For every hour take the median value for each component to create a similar file like that in **1.**

5. Start a second iteration for calculation of new solar quiet file like in **2.** now from the file generated in **4.**
Calculate mean values for every second from **4.** by interpolating.

6. Do the same as in **3.** but now from file generated in **5.**
Calculate new mean values by first doing a solar quiet interval file. That means get the time intervals that has values that are qualified to participate in the calculations of the new solar quiet file. Those intervals are produced by comparing the values of the file from **5.** with the value for the corresponding time of day for the date in question. The difference between those values must be within 10 nT (K-index 0) for that time to be qualified to participate in the calculation of the new solar quiet mean values. Observe that the difference now has been sharpened as compared with the calculations in **3.**

7. Do the same as **4.** but now from the file generated in **6.**
Calculate new mean values from the file that is generated from **6.** For every hour take the median value for each component to create a similar file like that in **1.**

8. The file generated in **7.** is now the new solar quiet file.

The variation of the solar quiet curve is less that 10 nT from month to month therefore the algorithm uses deviations from the sq curve of the previous month of 100 nT in the first iteration and 10 nT in the second iteration to calculate a new Solar Quiet curve.

Now that we have a solar quiet curve we can continue to make the K-indices calculations.

For every day in three hour intervals we calculate K-index.
That means for the following hour intervals:
0-3, 3-6, 5-9, 9-12, 12-15, 15-18, 18-21, 21-24
As Kiruna Magnetometer collects data every second the solar quiet curve with 24 values (one for each hour) is interpolated to get values for each second during a day.

In each interval we take the maximum deflection from solar quiet and the minimum deflection from solar quiet for the horizontal components (x and y) and subtract these values (max – min) to get a total deflection. The total deflection value determines the K-index value according to table 1. above.

Table 2: Example Solar Quiet file

X (pT)	Y (pT)	Z (pT)
10638894	*-91957*	*51817732*
10641294	*-92162*	*51817152*
10643882	*-92377*	*51817457*
10643891	*-92356*	*51817470*
10641757	*-92354*	*51818139*
10642735	*-93584*	*51817495*
10641265	*-95282*	*51817208*
10641714	*-95960*	*51814396*
10638799	*-94869*	*51817234*
10637983	*-95847*	*51817535*
10636815	*-99162*	*51818190*
10636732	*-99002*	*51818487*
10641875	*-93074*	*51812544*
10641863	*-93029*	*51812523*
10641310	*-98623*	*51820069*
10644902	*-97646*	*51816755*
10640593	*-91926*	*51807756*
10646601	*-96004*	*51818344*
10646277	*-95996*	*51818349*
10646277	*-95632*	*51817617*
10644188	*-94023*	*51816800*
10644110	*-90536*	*51810423*
10637333	*-91737*	*51806640*
10637400	*-91740*	*51806645*

Table 3: Example Solar Quiet Interval file

Starttime	Endtime
yyyymmddhh	yyyymmddhh
2016030223	2016030300
2016030317	2016030321
2016030418	2016030501
2016030502	2016030504
2016030600	2016030601
2016030603	2016030604
2016031302	2016031305
2016032217	2016032219
2016032316	2016032317
2016032517	2016032518
2016032521	2016032600
2016032616	2016032617
2016032816	2016032817
2016033122	2016033123